

Interaction-specific structure in coupled protein families: a modular, frozen-and-added Restricted Boltzmann Machine

Abstract

Many biological systems are organised not as single homogeneous statistical ensembles but as families of interacting subsystems: physically or functionally coupled protein families that have coevolved under a shared constraint. We are interested in modelling such interacting family pairs generatively, in the regime that is actually relevant for most biological interactions of interest: single-family sequence data are abundant, while paired, interaction-labelled data are scarce. We propose a modular Restricted Boltzmann Machine (RBM) architecture that exploits this asymmetry directly. Two RBMs are first pretrained independently on the abundant single-family data, until each reproduces its own family’s statistics; their parameters are then frozen inside a larger, paired model, and a small block of newly added hidden units, connected to the full joint visible layer, is left free to absorb whatever the two frozen, independent models fail to explain – by construction, only genuine inter-family structure. This isolates a minimal, identifiable channel for interaction-specific information, and makes the paired-training stage cheap in the one resource that is actually scarce. We validate this construction on a two-family Ising/Potts spin system with a known, hidden ground-truth coupling matrix, used as a controlled test case and confirmed stable across independent random realisations (Appendix A), and then apply the identical architecture and training recipe, unchanged, to two real pairs of coevolving Pfam protein domains that share a common family (Section 6). In both protein systems, the added units recover cross-family correlation structure that lies far above an independent-model null, though further from the finite-sample ceiling than in the controlled case; an independent, energy-based test correctly recovers the true interaction partner among candidate paralogs, well above chance, in both systems; and clamped conditional generation reproduces true-partner-specific structure beyond both a wrong-partner control and a naive consensus baseline. Taken together, these results support the central hypothesis – that interaction-specific structure can be attributed to a small, cheaply trainable channel added on top of frozen, independently pretrained models – while also indicating that real coevolutionary signal is harder to recover in full than the sparse, synthetic coupling of the controlled test case.

1 Motivation

Many complex systems of interest in statistical physics are not single homogeneous systems but collections of subsystems that interact weakly with one another: coupled spin lattices, interacting populations of neurons, or – the case that concerns us here – correlated families of biological sequences. Protein domain families evolve largely independently, subject to their own internal structural and functional constraints, but pairs of families that physically interact or act in the same pathway (a receptor and its ligand, a kinase and its substrate, two subunits of the same complex) are additionally constrained by the requirement of *mutual* compatibility: an amino-acid substitution on one side of an interaction interface is only tolerated if compensated, directly or diffusely, on the other side. This coevolutionary signal is exactly analogous to a genuine physical coupling between two otherwise weakly-interacting statistical-mechanical subsystems, and inferring it from sequence data alone is a classic, still largely open, inverse problem.

What makes this problem distinctive in practice is a strong data asymmetry. Single-family sequence databases are enormous and growing: hundreds of thousands of natural sequences are available for many individual Pfam domains. Data in which two specific, interacting family members are *paired* – known to co-occur in the same functional complex or operon – are comparatively scarce, since establishing a correct pairing typically requires either genomic proximity heuristics, experimental validation, or a species with a small enough paralog count that pairing is unambiguous. Any inference method that requires large amounts of paired data to work is, in this sense, solving the wrong problem: the practically relevant question is how much can be learned about the interaction itself from the much smaller paired data set, *given that the bulk of statistical structure – everything about each family in isolation – can already be learned essentially for free* from the abundant unpaired data.

This project’s central bet is to take that asymmetry seriously at the level of model architecture, not just training-set size. An RBM is already a natural generative model for a single sequence family: its hidden layer acts as an auxiliary field mediating effective, non-pairwise interactions among the visible (sequence) units once marginalised out, in the same spirit as a Hubbard–Stratonovich decoupling of a many-body interaction, which is what makes it more expressive than a purely pairwise (Potts/DCA-type) model while remaining tractable. Our proposal is to pretrain one such RBM per family on the abundant single-family data, *freeze* its parameters, and then ask whether a small, newly-added block of hidden units, connected to the full joint (paired) visible layer and trained only on the scarce paired data, is sufficient to capture the interaction-specific structure that a pair of independent, frozen models cannot represent by construction. If it is, the paired data requirement for learning interaction specificity is reduced to whatever is needed to fit a small number of new parameters, rather than an entire joint model.

This sits at the intersection of two questions, one practical and one more fundamental to representation learning:

- **Practical.** Can we build a generative model of an interacting protein family pair that is trainable with a realistically small paired data set, while still producing high-quality conditional and joint samples – ultimately, given a sequence from family A , a plausible, specific partner from family B ?
- **Fundamental.** What does a model learn, and how efficiently, when a small, structurally isolated subset of its parameters is forced to encode exactly the information that a set of frozen, pretrained components cannot – an explicit, testable instance of forcing a representation to specialise?

2 The scientific question

We restate the motivation of Section 1 as a precise, falsifiable question, so that the validation strategy of Section 5 can be judged against it directly.

Setting. Two protein families A and B , with N_A and N_B alignment columns respectively over a q -letter alphabet (amino acids plus gap, $q = 21$, for real sequences; a $q = 2$ Ising spin for the controlled test case of Appendix A). We have access to

- abundant single-family data sets D_A , D_B (unpaired sequences of each family separately, $|D_A|, |D_B|$ large), and
- a much smaller paired data set D_{AB} of sequences known, by some external criterion, to interact (e.g. genomic co-occurrence), $|D_{AB}| \ll |D_A|, |D_B|$.

Goal. Learn a joint generative model $p_\theta(v_A, v_B)$ such that:

- (a) the marginals $p_\theta(v_A)$ and $p_\theta(v_B)$ are as faithful to D_A and D_B as models trained directly and exclusively on D_A , D_B – pairing must not degrade what is already learnable from the abundant single-family data;
- (b) the joint distribution captures genuine inter-family structure – residue-residue covariation between sites of A and sites of B that reflects a true interaction constraint – learned efficiently from the small D_{AB} .

Falsifiable success criteria. Because there is generally no ground-truth coupling to compare against for real protein families, we adopt the same triangulation used throughout statistical inference from finite samples: any recovered signal must be judged relative to a *null* (what a model with no shared structure at all would produce) and a *ceiling* (what the finite paired sample itself can support, irrespective of the model). Concretely, the architecture of Section 3 is judged to be working if the inter-family structure recovered by the trained paired model

- lies far above the level recovered by two independently trained, uncoupled models sharing no hidden units at all (the *null*), and
- approaches, as closely as the amount of paired training data allows, the level of agreement between two independent finite samples of the true interacting distribution (the *ceiling*) –

while simultaneously satisfying (a): the frozen single-family blocks must still reproduce their own family’s statistics at equilibrium. A result in which the paired model sits at the null (no interaction signal learned) or in which pairing measurably corrupts single-family statistics would falsify the construction as currently designed.

3 Architecture: a modular, frozen-and-added RBM

An RBM is a pairwise energy-based model on a bipartite graph of visible units $\mathbf{v}$ and hidden units $\mathbf{h}$, with joint Boltzmann weight

$$p(\mathbf{v}, \mathbf{h}) \propto \exp[-E(\mathbf{v}, \mathbf{h})], \quad E(\mathbf{v}, \mathbf{h}) = -\mathbf{v}^\top W \mathbf{h} - a(\mathbf{v}) - b(\mathbf{h}), \quad (1)$$

where W is a visible–hidden weight matrix and a , b are layer-specific potentials. The absence of visible–visible and hidden–hidden couplings makes both conditionals, $p(\mathbf{h} \mid \mathbf{v})$ and $p(\mathbf{v} \mid \mathbf{h})$, factorised and hence efficiently sampled, while the marginalised hidden layer still mediates effective, non-pairwise interactions among the visible units (Section 1). The visible layer is Potts-encoded (q -state categorical units, one-hot, sampled through a Gumbel-softmax relaxation), paired with hidden units of the dReLU family (continuous, rectified, with both positive- and negative-going branches), which generalise the Bernoulli hidden unit to a richer response function; the same construction specialises to $q = 2$, Bernoulli-hidden “binary” RBMs as a limiting case, used in Appendix A for direct comparison. Both variants are wrapped with an online re-standardisation of the visible and hidden layers and, for the Potts layer, a zero-sum gauge fixing of the categorical weights.

3.1 The paired construction

Given two RBMs pretrained independently on D_A and D_B (Section 4), we build a single, larger *paired* model as follows (Fig. 1):

1. The visible layer of the paired model is the concatenation of the two families’ visible layers, $\mathbf{v} = (v_A, v_B)$, $N_A + N_B$ sites in total.

2. The hidden layer is the concatenation of three blocks: the n_A hidden units of the independently trained A -model, the n_B hidden units of the independently trained B -model, and n_{add} newly *added* hidden units with no prior identity.
3. The weight blocks $W_{v_A h_A}$ and $W_{v_B h_B}$ are initialised from the independently trained models and then *frozen*: after every gradient step of the paired model’s training, these blocks (and the corresponding visible/hidden biases) are reset to their pretrained reference values, discarding whatever update the optimiser proposed for them. This projection-based freezing is equivalent in effect to excluding these parameters from the gradient, but is implemented as a reset for convenience, and is checked at every logged step against drift.
4. The cross blocks $W_{v_B h_A}$ and $W_{v_A h_B}$ are pinned at exactly zero throughout training: the private hidden units of one family have, by construction, no direct view of the other family’s sequence.
5. The only parameters left genuinely free to adapt under gradient ascent are the weight blocks $W_{v_A h_{\text{add}}}$ and $W_{v_B h_{\text{add}}}$ connecting the added hidden units to the *full* joint visible layer, together with the added units’ own biases.

By construction, any cross-family statistical dependence that the paired model is capable of representing must flow through the added units: they are the sole channel connecting A and B . If they learn nothing, the paired model is exactly equivalent, in its predictions, to two independent, non-interacting RBMs – the null model of Section 2. If they learn something, it can only be inter-family structure, since every other parameter is already fixed to reproduce single-family statistics as well as the private models do. This is the sense in which the construction isolates a minimal, identifiable channel for interaction-specific information, and directly operationalises the fundamental question of Section 1: what is learned when only this channel is left free to specialise.

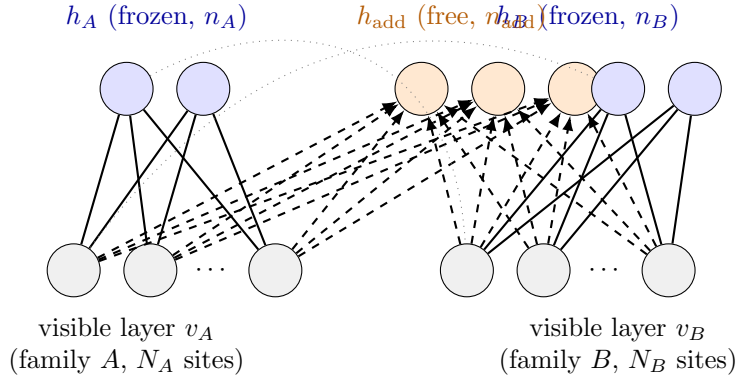


Figure 1: Architecture of the paired RBM. Solid edges: private A – A and B – B weight blocks, pretrained independently and reset to their reference values after every gradient step (“frozen”). Dotted grey edges: the cross blocks h_A – v_B and h_B – v_A , permanently pinned at zero. Dashed orange edges: the only free parameters in the paired model, connecting the n_{add} added hidden units to the full, joint visible layer.

4 Training curriculum

Maximum-likelihood training of an RBM requires the gradient of the log-likelihood, which contains a data-averaged and a model-averaged term; the latter is estimated by Persistent

Contrastive Divergence (PCD), i.e. a set of Monte Carlo chains that are carried over and advanced by a fixed number of sweeps between successive gradient steps, rather than reinitialised at every step. Training throughout uses PCD with the Adam adaptive step-size rule.

Training proceeds in two stages, deliberately separated so that the second, expensive-data stage is as cheap as possible in the number of free parameters and gradient steps required:

1. **Pretraining (abundant data).** Each family’s private RBM is trained independently on its own single-family data set (D_A or D_B) until its own held-out pseudolikelihood and moment-matching diagnostics (Section 5) indicate convergence, with no reference at all to the other family or to any paired data.
2. **Pair-training (scarce data).** The two pretrained models are assembled into the paired architecture of Section 3.1 and the added block is trained, with the private blocks frozen, on the paired data set D_{AB} only. Since the number of free parameters at this stage is a small fraction of the full paired model’s parameter count, and the private blocks already provide a correct single-family generative prior, this stage is expected to require substantially less data and substantially fewer gradient steps to converge than training the equivalent full joint model from scratch – this expectation is itself part of what Section 5 is designed to test.

5 Validation methodology

We use two complementary levels of diagnostic, applied identically to the controlled test case (Appendix A) and to the protein system (Section 6).

Training-time diagnostics. During pair-training we periodically compare, along the same short persistent Monte Carlo chains used for the gradient estimate, data-averaged and model-averaged statistics: visible–hidden correlations $\langle v_i h_\mu \rangle$ (“*vh-check*”), visible–visible correlations $\langle v_i v_j \rangle$ (“*vv-check*”), the inter-family correlation specifically (“*ab-check*”), hidden-unit firing rates from actual samples rather than mean-field values (to catch collapsed, always-on/off units that moment-based checks alone can miss), the weight norm of the added units’ incoming visible weights (to check they are acquiring receptive-field structure at all), and the raw free-gradient norm on the added block (to distinguish converged from merely stuck). A dedicated freeze check verifies, at every logged step, that the nominally frozen blocks have in fact not drifted from their reference values. These are cheap, on-line diagnostics of convergence, not a substitute for a proper equilibrium test, since the persistent chains used during training are advanced by only a handful of sweeps per step and are not guaranteed to be equilibrated.

Equilibrium validation. The decisive test is performed *after* training, on the frozen, converged model: a large number of independent Gibbs chains are run from random initial conditions for long enough that a free-energy burn-in diagnostic shows a stationary plateau (state-space size dependent – see the equilibration caveat below), and the resulting equilibrium samples are used to estimate the model’s cross-family correlation structure. This is compared against the best available truth proxy – the known coupling matrix in the controlled test case, a held-out paired split in the protein case – via (i) the fraction of the strongest true/reference couplings that are also among the strongest correlations recovered by the model (“top- K overlap”), and (ii) the correlation between the full recovered structure and the reference. Both quantities are also computed for the finite paired training sample itself (the *ceiling*, Section 2) and for the independent, uncoupled-model *null*. The same equilibrium samples are further decomposed by hidden block (frozen vs. added, mirroring Fig. 1) to check that pairing has

not corrupted the frozen blocks’ agreement with single-family data, and by intra-family correlation (“family preservation”) to check that recovering cross-family signal has not come at the cost of each family’s own statistics.

An equilibration caveat. The free-energy burn-in diagnostic is not a formality: the number of Gibbs sweeps needed to reach a stationary distribution grows sharply with the size of the state space, and a q^N -state categorical system with $q = 21$ and $N \sim 100$ –200 sites mixes far more slowly under single-site Gibbs updates than a $q = 2$, $N = 30$ -spin toy system. Equilibrium comparisons drawn from chains that have not visibly plateaued should be read as a lower, one-sided bound rather than a converged estimate, and this is checked explicitly, not assumed, for every system studied (see Appendix A for the controlled case and Section 6 for the protein case).

6 Application: real interacting protein-domain pairs

We apply the architecture and training curriculum of Sections 3–4, unchanged, to two real pairs of coevolving Pfam domains that share a common family *A*, PF00072 (response regulator receiver): a standard two-component signal-transduction pairing with PF00512 (histidine kinase), and a chemotaxis-pathway pairing with PF01339 (CheB methylesterase). Testing the construction on two independent family-*B* partners, with a common family-*A* pre-trained model family, lets us ask whether the recovered interaction signal is a property of the architecture or an accident of one particular pair. Family *A* is the first 111 alignment columns in both data sets; family *B* is the remaining columns, 66 for PF00512 and 174 for PF01339. Visible units are one-hot, $q = 21$ -state Potts variables (the 20 amino acids plus the alignment gap symbol). PF00072_PF00512_paired.fasta contains 77 880 sequence pairs; PF00072_PF01339_paired.faa contains 27 548. In both cases sequences are shuffled and split into training, validation and held-out sets, the held-out split serving as the truth proxy of Section 5 since no ground-truth coupling matrix exists for real sequence data.

6.1 Setup

The private RBMs use 150 (family *A*) and 100 (family *B*) hidden units, pretrained on the single-family alignment columns with the PCD/Adam recipe of Section 4. The paired model adds $n_{\text{add}} = 20$ free hidden units on top of the two frozen private blocks, connected to the full joint visible layer (177 sites for PF00512, 285 for PF01339), and is pair-trained on D_{AB} with the identical freeze-and-reset mechanism of Section 3.1. Cross-family coupling strength, at both the training-time and equilibrium stages, is measured by the Frobenius norm, over the $q \times q$ colour pairs, of the connected one-hot correlation tensor $C[a, i, b, j] = \langle v_i^a v_j^b \rangle - \langle v_i^a \rangle \langle v_j^b \rangle$ – amino-acid identities are not ordinal, so this replaces the simple ± 1 Pearson correlation used in the Ising case (Appendix A) with the standard, order-independent way of reducing a Potts model’s per-site-pair coupling to a single non-negative covariation score (cf. Frobenius-norm/APC scores in direct coupling analysis).

6.2 Training dynamics and freeze integrity

Table 1 summarises the private RBMs and the pair-training stage for the representative, most-trained configuration of each system. In both systems, training and validation pseudolikelihood track each other closely for both private RBMs (no overfitting), and the freeze check confirms the reset mechanism holds exactly, with zero drift, at every logged step throughout pair-training – as in the controlled test case. The training-time *ab*-check (measured on

short, non-equilibrated PCD chains, Section 5) rises steadily during pair-training in both systems, reaching $r \approx 0.34$ for PF00512 after 20 000 pair-training steps and $r \approx 0.62$ (train) / 0.72 (held-out) for PF01339 after 8000 steps – values well below the $r \approx 0.8$ reached at equilibrium-comparable training length on the synthetic Ising data (Appendix A), consistent with the slower mixing and larger state space expected for real protein alignments.

Table 1: Private-RBM fidelity and pair-training integrity, most-trained configuration per system ($N_{\text{HIDDEN},A} = 150$, $N_{\text{HIDDEN},B} = 100$, $n_{\text{add}} = 20$).

	PF00072–PF00512	PF00072–PF01339
RBM A: $\ell_{\text{train}} / \ell_{\text{val}}$	−1.137 / −1.193	−0.564 / −0.620
RBM B: $\ell_{\text{train}} / \ell_{\text{val}}$	−0.958 / −0.982	−0.718 / −0.725
freeze check	exact, every step	exact, every step
<i>ab</i> -check (training-time)	$r \approx 0.34$	$r \approx 0.62$

6.3 Equilibrium validation against a held-out reference

Table 2 reports the decisive, post-training test (Section 5): long Gibbs chains from the converged paired model, compared against the held-out reference, the training-data ceiling, and the independent-model null. In both systems the paired model clears the independent-model null by a wide margin – confirming that genuine, verifiable cross-family signal is learned on real sequence data, not only on the idealised synthetic model – but sits further from its own finite-sample ceiling than in the controlled test case (0.67–0.70 of the ceiling there, vs. roughly a third to a half of the ceiling here). The whole-matrix correlation carries a markedly higher null floor for PF01339 ($r = 0.539$) than for PF00512 ($r = 0.226$) or the controlled test case ($r \approx 0$), reflecting shared conservation patterns that two independent models can reproduce without any genuine coupling; the top- K overlap, which asks whether the *same* site pairs are identified as strongest, is the cleaner discriminator between systems and is consistently near zero for the null in both cases.

Table 2: Equilibrium validation against the held-out reference (5000 Gibbs samples, 5000 sweeps burn-in). “Train data” is the finite-sample ceiling; “independent” is the null model of two uncoupled, independently trained RBMs. Cf. Table 5 for the Ising analogue.

	paired model	train data (ceiling)	independent (null)
<i>PF00072–PF00512 (top-366 overlap)</i>			
top- K overlap	0.358	0.956	0.046
corr. with held-out	0.555	0.997	0.226
<i>PF00072–PF01339 (top-966 overlap)</i>			
top- K overlap	0.509	0.942	0.045
corr. with held-out	0.733	0.996	0.539

The same equilibrium samples, decomposed by hidden block, show the frozen private blocks reproducing single-family statistics well in both systems (PF00512: $\rho = 0.858$ for $\langle h_A v_A \rangle$, $\rho = 0.927$ for $\langle h_B v_B \rangle$; PF01339: $\rho = 0.925$ and $\rho = 0.856$ respectively), somewhat below the near-exact $\rho = 0.95$ –0.97 of the controlled test case but with no sign of systematic breakdown, while the added units track the full joint visible layer closely in both systems ($\rho = 0.887$ PF00512, $\rho = 0.938$ PF01339). Residual intra-family correlation shifts modestly between data and paired-model samples in both directions (PF00512: family A $0.035 \rightarrow 0.040$,

family *B* 0.061 $\rightarrow$ 0.038; PF01339: family *A* 0.056 $\rightarrow$ 0.044, family *B* 0.049 $\rightarrow$ 0.047) – small, non-systematic changes consistent with weak indirect coupling through the shared layer, not a breakdown of the frozen blocks (criterion (a) of Section 2).

6.4 True interaction-partner recovery

The correlation-matrix comparison above is an aggregate, population-level test. A complementary, and stricter, per-instance test is whether the trained paired model can identify which specific sequence of family *B* truly interacts with a given sequence of family *A*, among a pool of candidate paralogs from the same species. Since the free energy of the paired model decomposes as $F(v_A, v_B) = f_A(v_A) + f_B(v_B) + f_{\text{add}}(v_A, v_B)$ and only the last, added-unit term is non-separable in (v_A, v_B) , assignment by minimum joint free energy (via the Hungarian algorithm, within each species) is a clean test of what the added units alone have learned, uncontaminated by single-family energy scales.

Table 3 reports pooled and per-species accuracy across all qualifying species (those with two or more paralogs), together with a null-model and chance baseline, for both systems. In both cases recovery accuracy is far above chance and above the null model, and declines monotonically with the number of candidate paralogs per species (from above 90% at 2–3 candidates down to 16–20% at the largest paralog counts, always above the corresponding chance level). Because species contribute unequally redundant sequence data, we additionally report an accuracy reweighted by each pair’s effective sample size M_{eff} (inverse count of near-identical sequences at a 90% identity threshold), which corrects for a small number of near-clonal species otherwise dominating the per-species average; we take this reweighted figure as the primary accuracy estimate.

Table 3: True interaction-partner recovery among candidate paralogs (Hungarian assignment by minimum joint free energy, per species).

	species	pairs	pooled	per-species	M_{eff} -weighted	chance
PF00072–PF00512	1340	14 578	59.3%	74.7%	50.6%	17.4%
PF00072–PF01339	699	2 997	64.3%	84.0%	80.3%	36.6%

6.5 Generative quality and conditional specificity

For the PF00072–PF00512 system we further checked, on held-out sequences, whether the paired model generalises rather than memorises, and whether it produces genuinely partner-specific samples when conditioned on one family. Unconditional generation from the trained model covers 97.7% of the real-sequence manifold (a PCA-based coverage statistic) with 0% of generated sequences within 95% identity of any training example, and a mean nearest-training-neighbour identity of 56.1% – indicating genuine generalisation rather than memorisation of training sequences.

Conditional generation clamps a held-out sequence from one family and draws $R = 10$ replicas of the partner family, taking the majority-vote residue at each position; Table 4 compares the resulting reconstruction of the true partner’s identity against a wrong-partner control (a randomly chosen, non-interacting sequence from the same family) and a position-wise consensus baseline (uninformed by any specific partner). In both directions, majority-vote reconstruction of the true partner clearly exceeds both the wrong-partner control and the consensus baseline, with the largest relative margin over consensus at the least conserved alignment positions – precisely where a generic consensus carries no information and only genuine, partner-specific covariation can help.

Table 4: Clamped conditional generation, PF00072–PF00512, held-out sequences, majority vote over $R = 10$ replicas.

	$A \rightarrow B$	$B \rightarrow A$
true-partner recovery	55.1%	47.0%
consensus baseline	46.2%	40.4%
wrong-partner control	35.7%	33.5%

The same generative and conditional tests have not yet been repeated on the PF00072–PF01339 system; doing so, alongside a systematic M_{eff} -style redundancy correction analogous to Section 6.4, is the natural next step for this part of the study.

7 Discussion and outlook

The scientific question motivating this work – whether interaction-specific structure between coupled protein families can be attributed to a small, dedicated, cheaply-trainable block of hidden units, added on top of two frozen models pretrained on abundant single-family data alone – has a consistent, positive answer across every system studied. On the controlled two-family Ising/Potts spin system with a known, hidden inter-family coupling matrix, the construction recovers the large majority of the true coupling structure and, checked across three independent random realisations (Appendix A), does so reproducibly rather than by a fortunate draw. On two real, independent pairs of coevolving Pfam domains sharing a common family (Section 6), the same architecture and training recipe, unchanged, again recovers cross-family correlation structure clearly separated from the independent-model null, and this recovery is corroborated by two further, qualitatively different tests: a per-instance free-energy test correctly identifies the true interacting partner among candidate paralogs well above chance in both systems (Section 6.4), and clamped conditional generation reconstructs true-partner-specific structure beyond both a wrong-partner control and a naive consensus baseline, most clearly at the least conserved alignment positions (Section 6.5). That three independent lines of evidence – aggregate correlation recovery, per-instance partner identification, and conditional generation quality – agree is the strongest available support, short of direct structural validation, that the added units encode genuine interaction specificity rather than a correlational artefact of the fitting procedure.

The quantitative gap between the two protein systems and the controlled test case is real and worth stating plainly: recovered top- K overlap reaches only a third to a half of its own finite-sample ceiling on real data, against roughly two-thirds to three-quarters of ceiling in the controlled case. Two features of the controlled study anticipated this gap and remain the most likely explanations: a much larger, $q = 21$ categorical state space mixes more slowly under Gibbs sampling than the 30-spin toy system, so real equilibrium estimates may still carry some residual bias even at the chain lengths used here; and real coevolutionary coupling is plausibly denser and more structured than the sparse, uniformly-random synthetic matrix of Appendix A, which is a harder target for a fixed, small n_{add} . That the true-pair and conditional-generation tests both nonetheless recover a strong, verifiable signal indicates the shortfall is one of *completeness* – the added units have not yet captured the full extent of the true coupling – rather than of validity: what is recovered already behaves as genuine interaction-specific structure by two independent measures.

Open items follow directly from this picture. On the protein side: extend the generative and conditional-generation tests of Section 6.5 to the PF00072–PF01339 system; apply the M_{eff} -style redundancy correction of Section 6.4 systematically wherever species or sequence

redundancy could bias an accuracy estimate; and, guided by the two confounds above, scan n_{add} and Gibbs chain length to determine how much of the remaining gap to the ceiling closes with more capacity versus more sampling. On the controlled side, the seed-averaging caveat of earlier iterations of this study has been closed (Appendix A); the natural remaining extension is a systematic scan over inter-family coupling density and strength, to map where the freeze-and-add construction stops being sufficient and a fuller joint retraining becomes necessary. Longer term, and building directly on Section 6.4, the same free-energy decomposition that identifies true partners among candidates can be turned into a practical pairing-assignment tool for genuinely unpaired sequence data, connecting this work to a related line of investigation on pretrained, frozen-and-added RBMs for protein interaction specificity.

A A controlled test case: the two-family Ising model

This appendix specialises the general architecture and validation methodology of Sections 3–5 to a toy statistical-mechanics system with a known, hidden ground truth, used to establish the proof of principle summarised in Section 7. Unlike Section 6, results here are complete.

A.1 Hamiltonian and equilibrium sampling

Each family $X \in \{A, B\}$ consists of $N = 30$ Ising spins $s_i^X = \pm 1$, $i = 1, \dots, N$ (i.e. $q = 2$ in the notation of Section 2). The joint system of $2N = 60$ spins is governed by the Hamiltonian

$$\mathcal{H}(\mathbf{s}_A, \mathbf{s}_B) = - \sum_{i < j} J_{ij}^A s_i^A s_j^A - \sum_{i < j} J_{ij}^B s_i^B s_j^B - \sum_{i, j} K_{ij} s_i^A s_j^B - \sum_i h_i^A s_i^A - \sum_i h_i^B s_i^B, \quad (2)$$

sampled at inverse temperature $\beta = 0.5$. The intra-family couplings J^A and J^B are independent sparse random-bond graphs (edge density 0.12, $J \in [-0.8, 0.8]$, fields $h \in [-0.05, 0.05]$) – an ordinary dilute Sherrington–Kirkpatrick-type random-bond Ising model for each family separately, chosen only for concreteness. The inter-family coupling matrix $K \in \mathbb{R}^{30 \times 30}$ is the ground-truth object of interest: a sparse matrix with a fixed number n_{links} of non-zero (i, j) pairs, each $K_{ij} \in [-0.9, 0.9]$, chosen at random, all other entries exactly zero. We generated data at $n_{\text{links}} = 20$ (2.2% of the 900 possible cross pairs) and $n_{\text{links}} = 100$ (11.1%); the sparser set sits close to the sampling-noise floor discussed below, so the denser set is used for the headline results.

Samples are drawn from $p(\mathbf{s}_A, \mathbf{s}_B) \propto e^{-\beta \mathcal{H}}$ by single-spin-flip Metropolis–Hastings, 5000 sweeps of burn-in and 100-sweep thinning, 20 000 total joint configurations, split 70/15/15 into training, validation and test sets. No information about J^A , J^B or K is passed to the learning stage.

A.2 Training and the paired model

Each private RBM (family A , family B) has 30 hidden units and is pretrained for 10 000 PCD/Adam steps ($K = 50$ sweeps/step, minibatch 512), following Section 4. The paired model, built as in Section 3.1 with $n_{\text{add}} = 30$ added units, is then pair-trained for a further 3000 steps on the joint (here, fully synthetic and hence unlimited) data. Results are reported for both the Potts+dReLU and the Binary variant of the architecture.

A.3 An important caveat on sparsity

Because K is very sparse (at most 100 non-zero entries out of 900), the whole-matrix Pearson correlation between any noisy estimate of the cross-correlation matrix and K is structurally bounded well below 1, even for a perfect model: correlating a mostly-zero signal against a

dense, noisy estimate dilutes the true signal with sampling noise on all the entries that are exactly zero by construction. Restricting the same comparison to only the 100 truly non-zero entries of K raises the correlation from $r = 0.824$ to $r = 0.992$ using the training data alone. The relevant benchmark is therefore not $r = 1$ but the ceiling set by the finite-sample data themselves, against which the paired model is judged throughout (Section 2).

A.4 Results

A.4.1 Individual RBM fidelity and training dynamics

Before pairing, each private RBM is checked against held-out data. Pseudo-likelihood on training and validation sets track each other closely for both families (Potts: $\ell = -0.598$ train vs. -0.592 validation for family A ; Binary: -0.598 vs. -0.598), indicating no overfitting at this training length (Fig. 2). The vh -check ($r = 0.94$ Potts, $r = 0.87$ Binary) is systematically higher than the harder vv -check ($r = 0.82$ both), as expected: reproducing hidden-unit activation statistics is a softer target than reproducing the full pairwise correlation structure of the spins themselves.

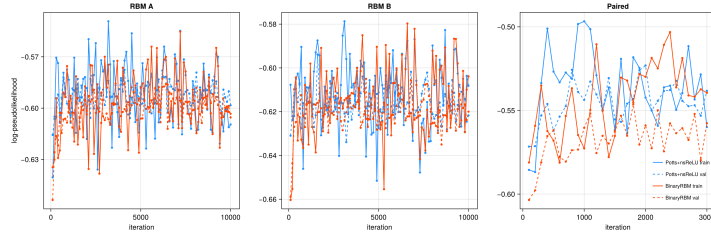


Figure 2: Training and validation pseudo-likelihood of the private RBMs (family A , family B) and of the paired RBM, as a function of gradient step, Potts (top) and Binary (bottom) variants.

Once the paired model is assembled, the freeze check confirms the private blocks remain exactly at their reference values throughout (30/30 frozen parameters unchanged at every logged step, both variants). The added units’ own vh -correlation grows steadily over the 3000 pair-training steps (Fig. 3) and their weight norm increases by more than an order of magnitude ($\times 14.9$ Potts, $\times 9.7$ Binary, from initialisation), with a non-negligible gradient remaining at the end of training and no unit collapsed to a dead, always-on or always-off state.

A.4.2 Equilibrium validation against the ground truth

Fig. 4 confirms 5000 sweeps is sufficient for 1000 independent Gibbs chains to reach a stationary free-energy plateau before equilibrium samples are recorded.

Fig. 5 decomposes equilibrium samples by hidden block, mirroring Fig. 1: the frozen A – A and B – B blocks reproduce the data almost exactly ($\rho = 0.95, 0.97$), confirming pairing does not disturb what the private models had already learned, while the added units’ correlation with the full joint visible layer tracks the data-side estimate at $\rho = 0.997$.

The central result is the direct comparison, at equilibrium, between the model’s recovered cross-family correlation matrix and the true coupling matrix K (Fig. 6, Table 5). The paired model recovers 67% (Potts) and 65% (Binary) of the 100 strongest true couplings among its own 100 strongest recovered correlations, against 74% for the training data themselves (the finite-sample ceiling) and 6–8% for two independent, uncoupled RBMs (essentially chance level). The whole-matrix correlation against K reaches $r = 0.796$ (Potts) and $r = 0.786$

(Binary), against a data ceiling of $r = 0.844$ and an independent-model baseline consistent with zero ($r = 0.012$ Potts, $r = -0.039$ Binary).

Table 5: Equilibrium validation against the true coupling matrix K (Gibbs samples, 1000 chains, 5000 sweeps burn-in). “Data” is the finite-sample ceiling; “independent” is the null model of two uncoupled, independently trained RBMs.

	paired model	data (ceiling)	independent (null)
<i>Potts + dReLU</i>			
top-100 overlap	0.67	0.74	0.06
corr. with K	0.796	0.844	0.012
<i>Binary</i>			
top-100 overlap	0.65	0.74	0.08
corr. with K	0.786	0.844	-0.039

Finally, recovering cross-family structure does not come at the cost of corrupting each family’s own private statistics (Fig. 7): residual off-diagonal intra-family correlation, sampled from the full paired model, stays close to its data value (family A : $0.057 \rightarrow 0.068$; family B : $0.048 \rightarrow 0.058/0.060$), a small degradation consistent with weak, indirect intra-family coupling introduced through the shared hidden layer, not with any breakdown of the frozen blocks.

A.4.3 Robustness across random seeds

The result above uses a single realisation of the random intra- and inter-family couplings and a single training run. To check this is not a fortunate draw, we repeated the full pretrain–freeze–pair–train pipeline at three independent random seeds, on the strong-field variant of the data set (local fields $h \in [-0.3, 0.3]$, introduced to keep the per-site marginal signal well clear of sampling noise) with $n_{\text{add}} = 40$ added units. Table 6 reports the range across seeds of the same headline quantities as Table 5.

Table 6: Equilibrium validation across three random seeds, strong-field data set, $n_{\text{add}} = 40$ (5000 Gibbs samples, 5000 sweeps burn-in each). Range given as (min–max) across the three seeds.

	paired model (range)	data (ceiling)	independent (null, range)
top-100 overlap	0.68–0.70	0.72	0.09–0.12
corr. with K	0.818–0.837	0.845	-0.04–0.05

The freeze check passes exactly, with zero drift, at every seed; private RBM pseudolikelihoods and vh -correlations ($r \approx 0.99$) are likewise consistent across seeds. The spread across seeds is small relative to the gap between the paired model and the null, confirming that the qualitative recovery reported above is a stable property of the construction in this setting, not an artefact of one particular random draw.

A.4.4 Potts versus Binary

The Potts+dReLU variant performs modestly but consistently better than the Binary RBM on training-time diagnostics (private-model vh -correlation 0.94 vs. 0.87; added-unit equilibrium tracking $\rho = 0.997$ vs. 0.81), yet this advantage barely shows up in the decisive test against

ground truth (top- K overlap 0.67 vs. 0.65; correlation with K 0.796 vs. 0.786). The categorical representation appears to yield cleaner internal statistics, without materially changing how much genuine cross-family signal is ultimately extracted in this regime.

A.5 Discussion

A small, dedicated block of hidden units, added on top of two frozen, independently pretrained models, recovers the large majority of an otherwise hidden inter-family coupling structure, confirmed directly against known ground truth by equilibrium Monte Carlo sampling rather than inferred indirectly from training curves. Two caveats: the density of the inter-family coupling matters (the sparser, $n_{\text{links}} = 20$ data set still clears the independent-model baseline, but with a noticeably weaker absolute signal, indicating very sparse, weak cross-couplings are harder to disentangle from sampling noise at fixed training length); and the natural whole-matrix correlation against a sparse ground truth carries a structural ceiling well below 1 (Section A.3), making the finite-sample ceiling, not $r = 1$, the correct benchmark throughout.

This remains a controlled, minimal test bed, validated at its headline configuration and, separately, confirmed stable across three random seeds at a second configuration (Section A.4.3); a systematic scan of β , hidden-layer size, or training length was not performed. A natural next step, orthogonal to the real-data question of Section 6, is a systematic scan over n_{add} and over inter-family coupling density and strength, extending the single-configuration headline result and the seed-robustness check into a full map of where the freeze-and-add construction stops being sufficient and a fuller joint retraining becomes necessary.

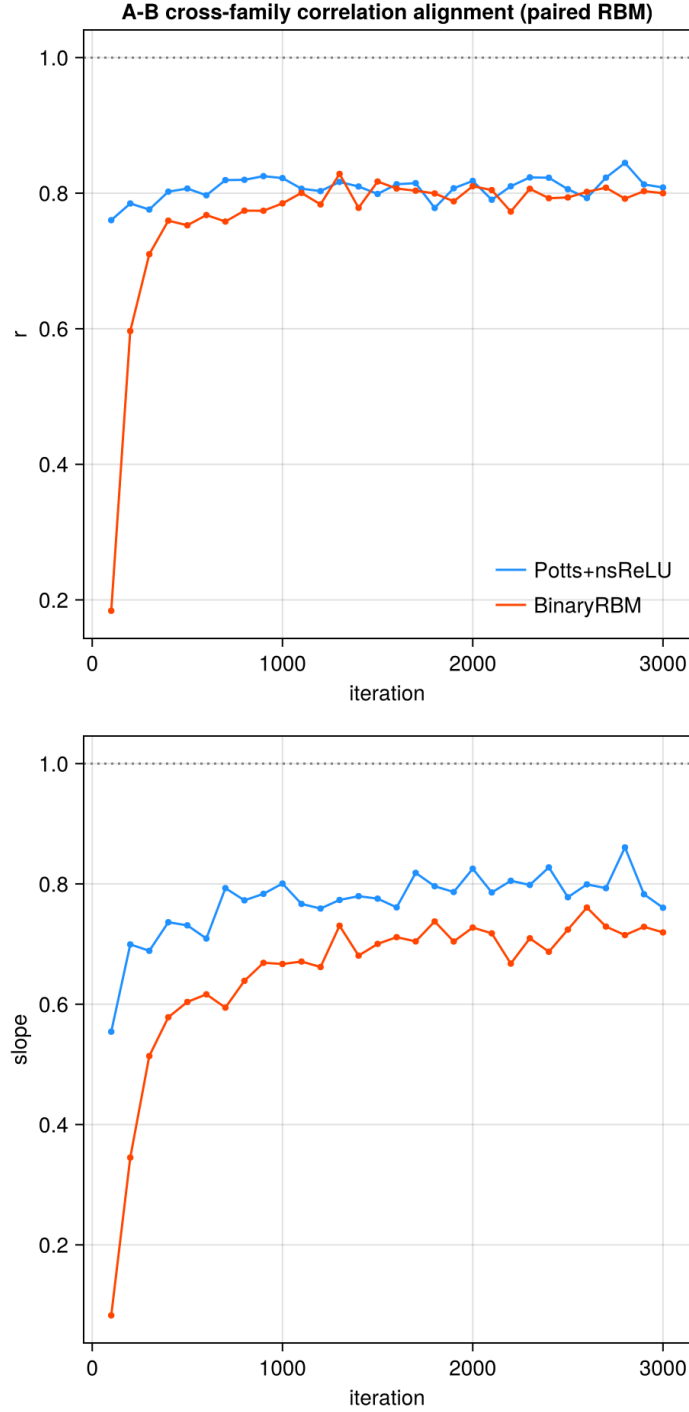


Figure 3: Evolution, over the 3000 pair-training steps, of the Pearson correlation between data-side and model-side inter-family correlations $\langle v_i^A v_j^B \rangle$ (the “ ab -check”), Potts and Binary paired models. Both climb from a poorly correlated initial state ($K_{AB} = 0$, independent-model regime) towards $r \approx 0.8$.

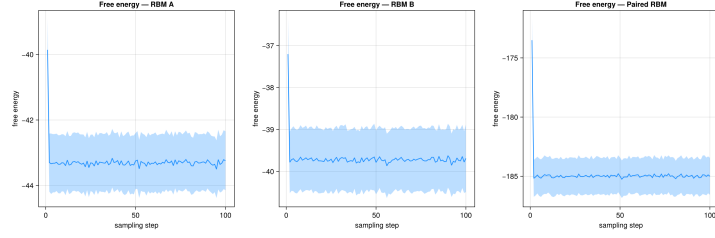


Figure 4: Free energy of the Gibbs chains as a function of the number of Monte Carlo sweeps, averaged over 1000 independent chains, for the private and paired Potts models. A stationary plateau is reached well before the 5000-sweep burn-in used throughout.

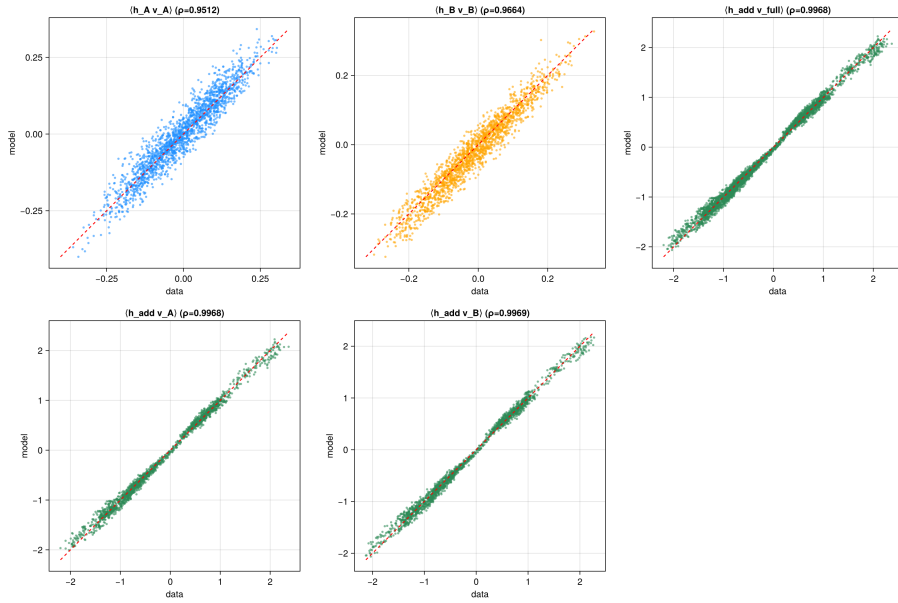


Figure 5: Data- vs. model-side visible–hidden correlations $\langle vh \rangle$, from equilibrium Gibbs samples of the Potts paired model, split by hidden block: private A – A and B – B blocks (frozen, should reproduce the data essentially exactly), and the added units against the full joint visible layer (the only block that can encode genuinely new, cross-family information).

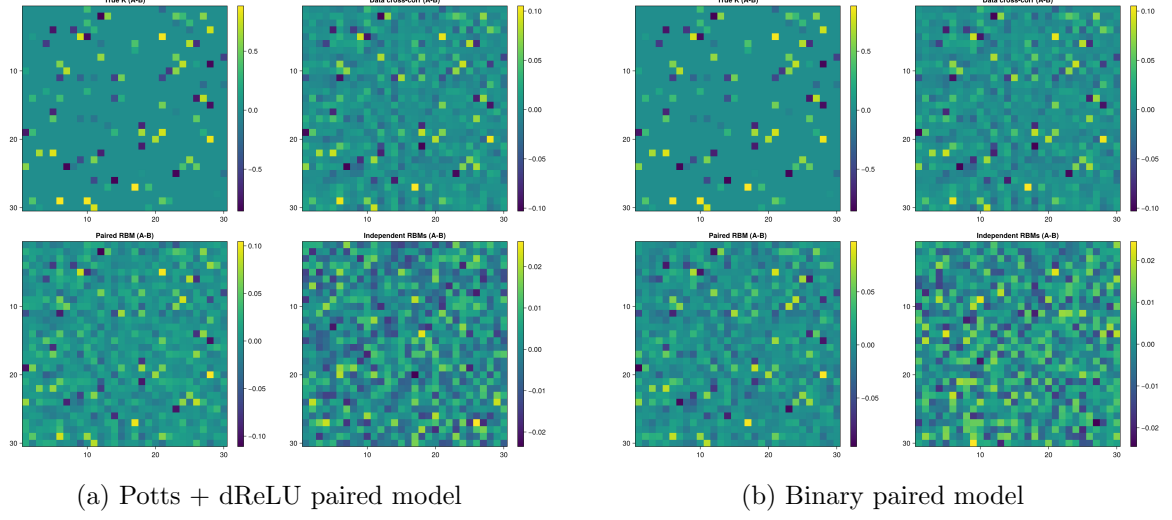


Figure 6: Cross-family correlation matrices at equilibrium: true coupling K , empirical training-data correlation, correlation recovered by the paired model, and correlation obtained from two independent, uncoupled RBMs. The paired model visibly reproduces the sparse support of K that is absent from the independent-model baseline.

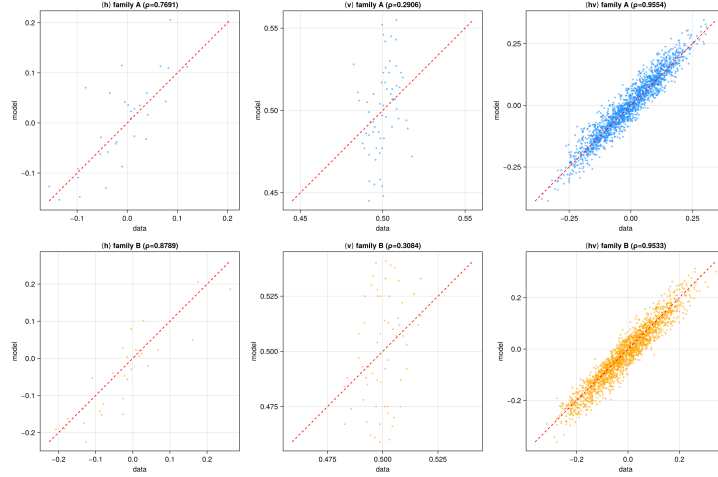


Figure 7: Intra-family correlation matrices, data vs. paired-model equilibrium samples, family A and family B separately. The frozen blocks preserve the private structure learned before pairing, up to a small perturbation induced by the shared added units.